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IMPLEMENT PACKET SNIFFING USING RAW SOCKETS IN PYTHON

AIM:

To capture and inspect network packets on a local interface using Python raw sockets — for learning or authorized troubleshooting (show source/destination MAC/IP, protocol, and ports).

PROCEDURE:

1. Confirm you have permission to sniff the target network.
2. Open a raw socket at the link layer (AF_PACKET, SOCK_RAW) to receive frames.
3. Bind the socket to the interface you want to monitor (e.g., eth0).
4. Loop reading packets from the socket.
5. Parse minimal headers (Ethernet → IPv4 → TCP/UDP) to extract addresses/ports.
6. Print/store/analyze the fields you care about.
7. Close the socket and stop when finished.

PROGRAM:

```
import socket
import struct
import binascii
import textwrap
```

```
def main():
```

```
host = socket.gethostbyname(socket.gethostname())
```

```
print('IP: {}'.format(host))
```

```
conn = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
```

```
conn.bind((host, 0))
```

```
conn.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
```

```
conn.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)
```

```
while True:
```

```
    raw_data, addr = conn.recvfrom(65536)
```

```
    dest_mac, src_mac, eth_proto, data = ethernet_frame(raw_data)
```

```
    print('\nEthernet Frame:')
```

```
    print("Destination MAC: {}".format(dest_mac))
```

```
    print("Source MAC: {}".format(src_mac))
```

```
    print("Protocol: {}".format(eth_proto))
```

```
def ethernet_frame(data):
```

```
    dest_mac, src_mac, proto = struct.unpack('!6s6s2s', data[:14])
```

```
    return get_mac_addr(dest_mac), get_mac_addr(src_mac),  
    get_protocol(proto), data[14:]
```

```
def get_mac_addr(bytes_addr):
```

```
    bytes_str = map('{:02x}'.format, bytes_addr)
```

```
mac_address = ':'.join(bytes_str).upper()
```

```
return mac_address
```

```
def get_protocol(bytes_proto):
```

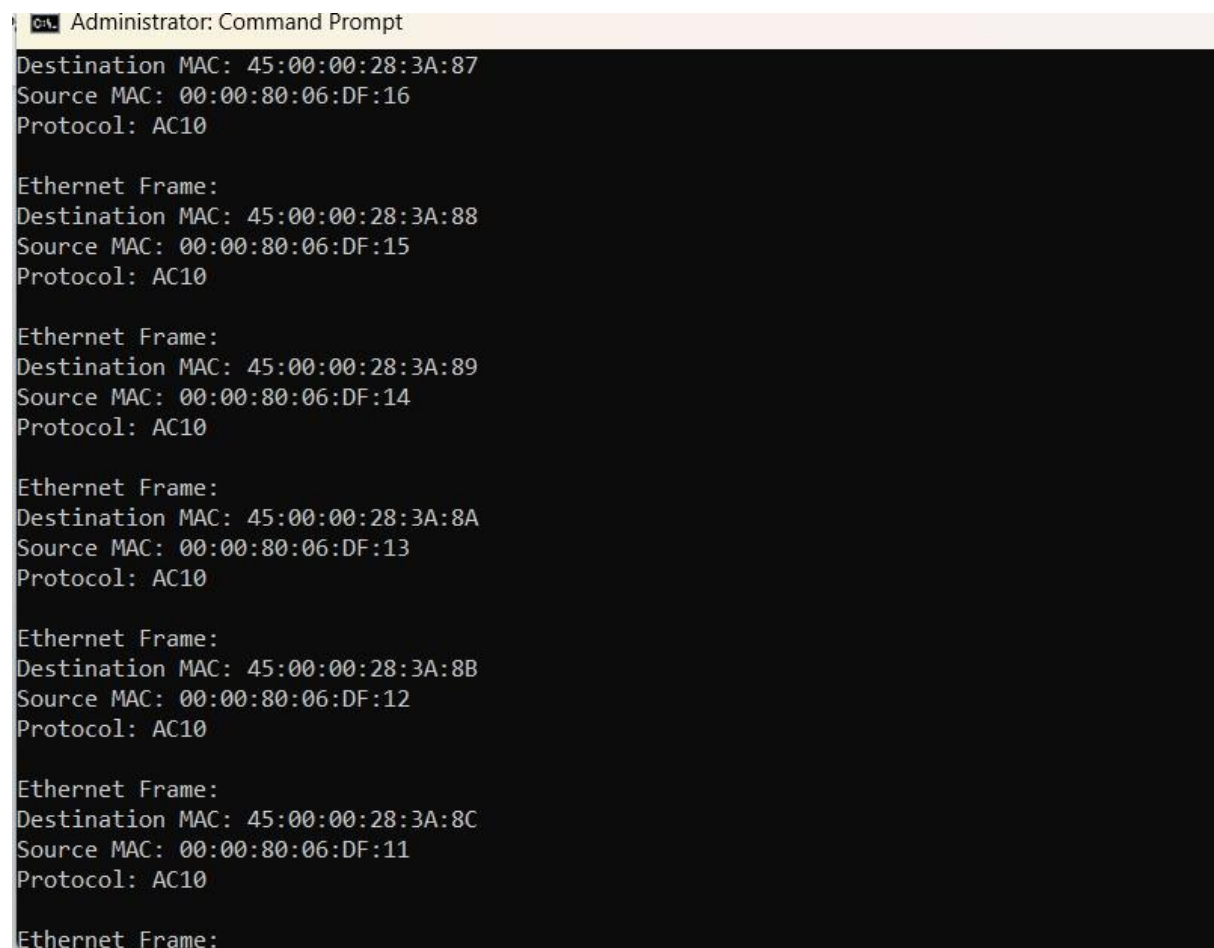
```
    bytes_str = map('{:02x}'.format, bytes_proto)
```

```
    protocol = ''.join(bytes_str).upper()
```

```
    return protocol
```

```
main()
```

OUTPUT:



```
Administrator: Command Prompt
Destination MAC: 45:00:00:28:3A:87
Source MAC: 00:00:80:06:DF:16
Protocol: AC10

Ethernet Frame:
Destination MAC: 45:00:00:28:3A:88
Source MAC: 00:00:80:06:DF:15
Protocol: AC10

Ethernet Frame:
Destination MAC: 45:00:00:28:3A:89
Source MAC: 00:00:80:06:DF:14
Protocol: AC10

Ethernet Frame:
Destination MAC: 45:00:00:28:3A:8A
Source MAC: 00:00:80:06:DF:13
Protocol: AC10

Ethernet Frame:
Destination MAC: 45:00:00:28:3A:8B
Source MAC: 00:00:80:06:DF:12
Protocol: AC10

Ethernet Frame:
Destination MAC: 45:00:00:28:3A:8C
Source MAC: 00:00:80:06:DF:11
Protocol: AC10

Ethernet Frame:
```

RESULT:

The program shows the source and destination MAC address and protocol of network packets it captures.